Geometry & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Rational exponents

Consider the function $f(x) = 3x^2 - 2x + 5$. Find the value of $f'(1)$.

2 [Foundational] • Target: Exponential functions

A circle with centre O has radius $r = 8$ cm. Chord AB subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc AB. (b) Find the exact area of the sector OAB. (c) Find the exact area of the minor segment enclosed between chord AB and arc AB.

3 [Foundational] • Target: Applications of exponential functions

Consider the function $f(x) = e^{2x} - 4e^x + 3$. Let $u = e^x$. Find the minimum value of the function in terms of u.

4 [Foundational] • Target: The natural exponent

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

5 [Foundational] • Target: Logarithms

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

6 [Foundational] • Target: Logarithm laws and change of base

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

7 [Foundational] • Target: Rational exponents

Points $A(1, 5)$ and $B(-2, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Exponential functions

Points $A(-2, 3)$ and $B(-3, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

9 [Intermediate] • Target: Applications of exponential functions

Points $A(2, -1)$ and $B(7, 5)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

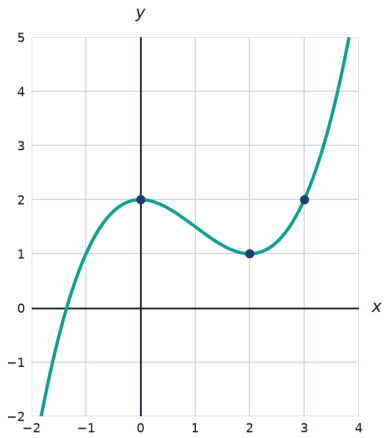
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10 [Intermediate] • Target: The natural exponent

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

11 [Intermediate] • Target: Applications of exponential functions

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(5)$.

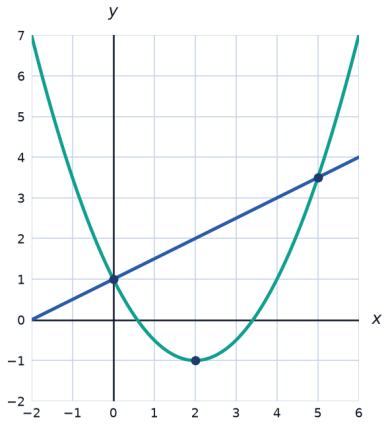


12 [Intermediate] • Target: Applications of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

13 [Intermediate] • Target: The natural exponent

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(2)$.



14 [Intermediate] • Target: The natural exponent

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: The natural exponent • Cross-Lesson: 2.01 Functions

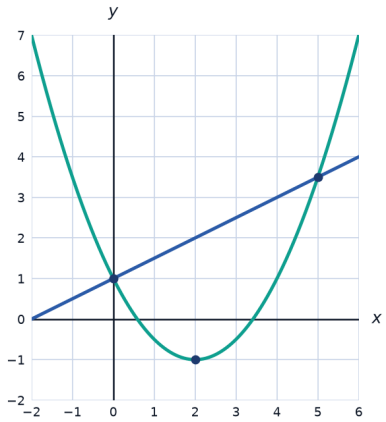
A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

16 [Challenging] • Target: The natural exponent • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(3)$.

17 [Challenging] • Target: The natural exponent • Cross-Lesson: 5.01 Calculus

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



18 [Challenging] • Target: The natural exponent • Cross-Lesson: 6.01 AI SL — Specific Topics

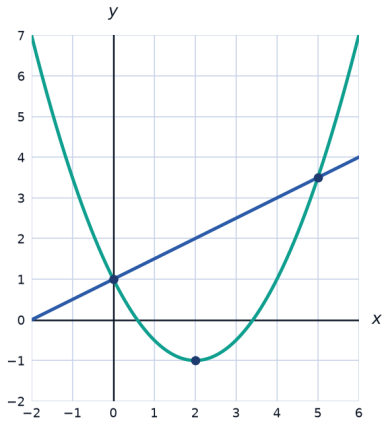
For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(4)$.

19 [Challenging] • Target: The natural exponent • Cross-Lesson: 1.01 Number & Algebra

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

20 [Challenging] • Target: The natural exponent • Cross-Lesson: 2.01 Functions

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(5)$.

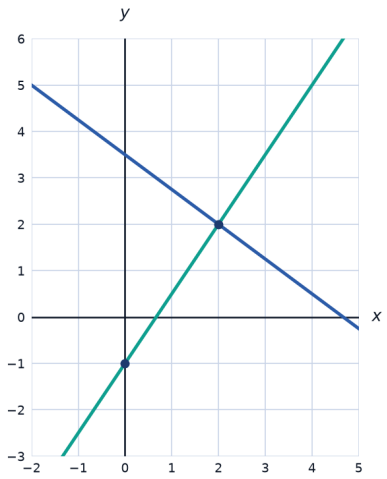


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Rational exponents

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(2)$.



a. 14

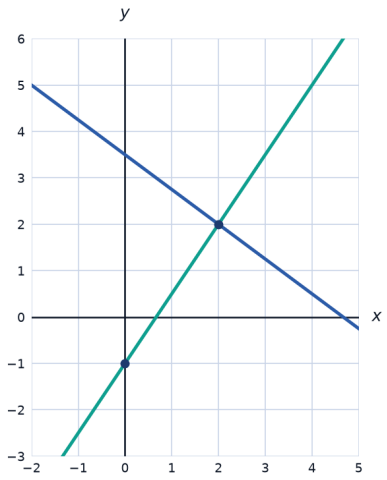
b. 18

c. 8

d. 18

22 [Foundational] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

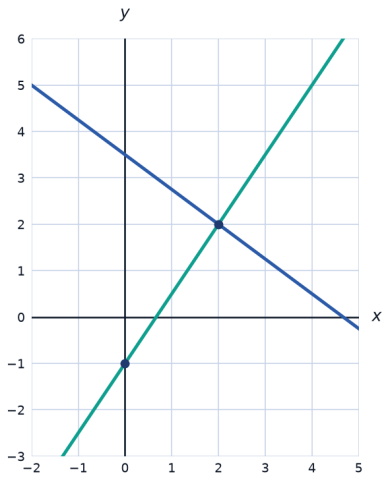
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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23 [Foundational] • Target: Rational exponents

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(5)$.



a. 26

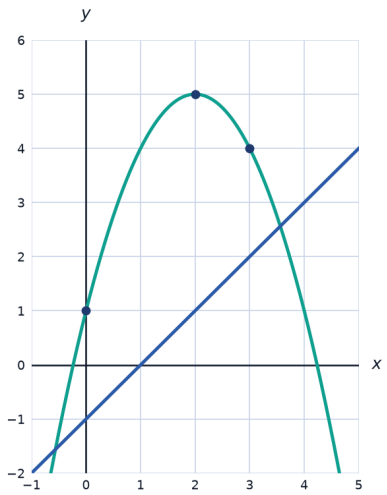
b. 30

c. 20

d. 30

24 [Foundational] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(2)$.



a. 17

b. 22

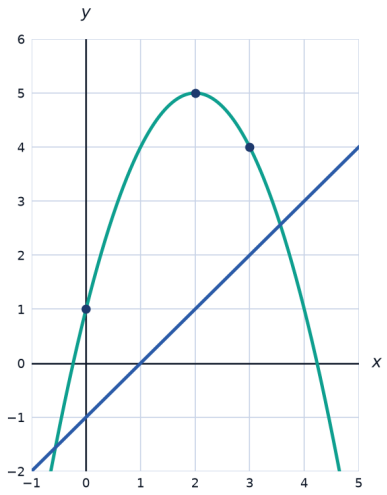
c. 10

d. 21

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25 [Foundational] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

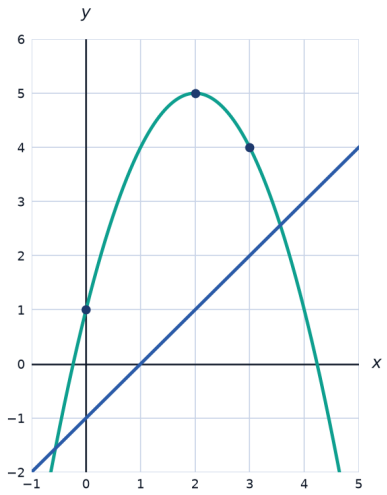
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

26 [Foundational] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(5)$.



a. 32

b. 37

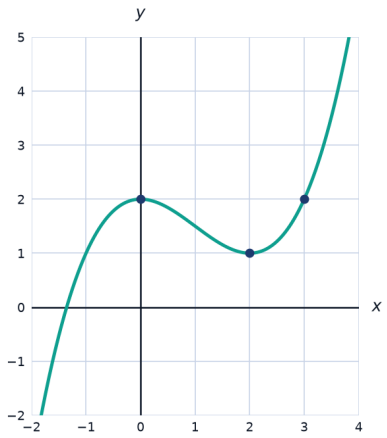
c. 25

d. 36

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27 [Foundational] • Target: Applications of exponential functions

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(2)$.



a. 20

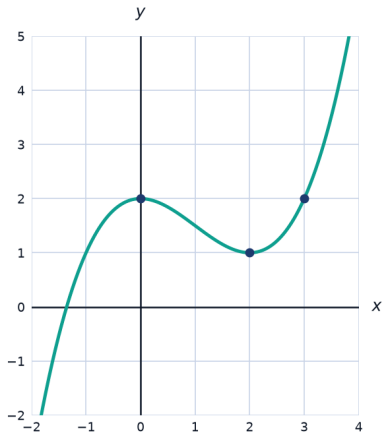
b. 26

c. 12

d. 24

28 [Foundational] • Target: Applications of exponential functions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Rational exponents

A data set recorded for rational exponents consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Rational exponents

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(3)$.

a. 18

b. 22

c. 12

d. 22

32 [Intermediate] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

33 [Intermediate] • Target: Rational exponents

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

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34 [Intermediate] • Target: Exponential functions

A data set recorded for exponential functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

35 [Intermediate] • Target: Exponential functions

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(3)$.

a. 22

b. 27

c. 15

d. 26

36 [Intermediate] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

37 [Intermediate] • Target: Exponential functions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 6.01 AI SL — Specific Topics

A data set recorded for applications of exponential functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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39 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 1.01 Number & Algebra

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(3)$.

a. 26

b. 32

c. 18

d. 30

40 [Challenging] • Target: Applications of exponential functions • Cross-Lesson: 2.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

41 [Challenging] • Target: The natural exponent • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

42 [Challenging] • Target: Logarithms • Cross-Lesson: 5.01 Calculus

A data set recorded for logarithms consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

43 [Challenging] • Target: Logarithms • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 3x + 3$ in logarithms, evaluate $f(3)$.

a. 12

b. 15

c. 9

d. 16

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44 [Challenging] • Target: Logarithm laws and change of base • Cross-Lesson: 1.01 Number & Algebra

A data set recorded for logarithm laws and change of base consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

45 [Challenging] • Target: Logarithm laws and change of base • Cross-Lesson: 2.01 Functions

For the model $f(x) = 4x + 4$ in logarithm laws and change of base, evaluate $f(3)$.

a. 16

b. 20

c. 12

d. 20

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